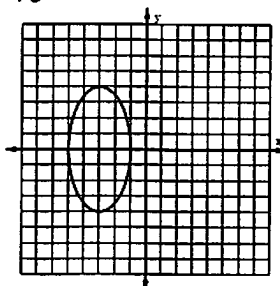


Name: _____		Date: _____	
Topic: _____		Class: _____	

Main Ideas/Questions	Notes/Examples													
TYPES OF SYMMETRY	Line Symmetry The graph can be folded along a line so that the two halves match.	Point Symmetry The graph remains unchanged when rotated 180° about a point.												
	The <u>x - axis</u> and <u>y - axis</u> are common lines of symmetry. The <u>origin</u> is a common point of symmetry.													
	The following graphical and algebraic tests can be used to determine if the graph of a relation is symmetric to the x-axis, y-axis, and/or origin. <table border="1" style="width: 100%; border-collapse: collapse;"> <thead> <tr> <th style="width: 10%;"></th> <th style="width: 45%; text-align: center;">Graphical Test</th> <th style="width: 45%; text-align: center;">Algebraic Test</th> </tr> </thead> <tbody> <tr> <td style="text-align: center; vertical-align: middle;">x-axis</td> <td style="padding: 5px;">For every point (x, y) on the graph, the point $(x, -y)$ is also on the graph.</td> <td style="padding: 5px;">Replacing <u>y</u> with <u>$-y$</u> results in equivalent equations.</td> </tr> <tr> <td style="text-align: center; vertical-align: middle;">y-axis</td> <td style="padding: 5px;">For every point (x, y) on the graph, the point $(-x, y)$ is also on the graph.</td> <td style="padding: 5px;">Replacing <u>x</u> with <u>$-x$</u> results in equivalent equations.</td> </tr> <tr> <td style="text-align: center; vertical-align: middle;">origin</td> <td style="padding: 5px;">For every point (x, y) on the graph, the point $(-x, -y)$ is also on the graph.</td> <td style="padding: 5px;">Replacing <u>y</u> with <u>$-y$</u> AND <u>x</u> with <u>$-x$</u> results in equivalent equations.</td> </tr> </tbody> </table>				Graphical Test	Algebraic Test	x-axis	For every point (x, y) on the graph, the point $(x, -y)$ is also on the graph.	Replacing <u>y</u> with <u>$-y$</u> results in equivalent equations.	y-axis	For every point (x, y) on the graph, the point $(-x, y)$ is also on the graph.	Replacing <u>x</u> with <u>$-x$</u> results in equivalent equations.	origin	For every point (x, y) on the graph, the point $(-x, -y)$ is also on the graph.
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TESTS FOR SYMMETRY														
	Directions: Use the graph to determine if the relation is symmetrical to the x-axis, y-axis, and/or origin. Confirm your answer algebraically. 1. $y = x^4 - 4x^2$ ☐ x-axis ☒ y-axis ☐ origin ☐ none $y = (-x)^4 - 4(-x)^2$ $y = x^4 - 4x^2 \quad \checkmark$ 2. $xy = 6$ ☐ x-axis ☐ y-axis ☒ origin ☐ none $(-x)(-y) = 6$ $xy = 6 \quad \checkmark$													
Examples														

3. $4(x+3)^2 + y^2 = 16$

- ☒ x-axis
☐ y-axis
☐ origin
☐ none

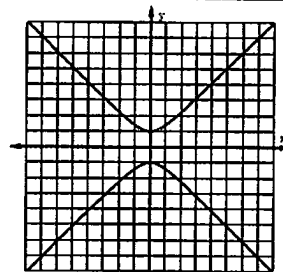


$$4(x+3)^2 + (-y)^2 = 16$$

$$4(x+3)^2 + y^2 = 16 \quad \checkmark$$

4. $y^2 - x^2 = 1$

- ☒ x-axis
☒ y-axis
☒ origin
☐ none



$$(-y)^2 - x^2 = 1$$

$$y^2 - x^2 = 1 \quad \checkmark$$

$$y^2 - (-x)^2 = 1$$

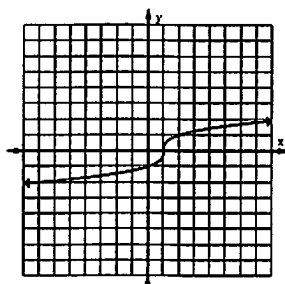
$$y^2 - x^2 = 1 \quad \checkmark$$

$$(-y)^2 - (-x)^2 = 1$$

$$y^2 - x^2 = 1 \quad \checkmark$$

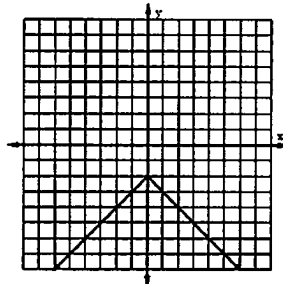
5. $y = \sqrt[3]{x-1}$

- ☐ x-axis
☐ y-axis
☐ origin
☒ none



6. $y = -|x| - 2$

- ☐ x-axis
☒ y-axis
☐ origin
☐ none



$$y = -|-x| - 2$$

$$y = -x - 2 \quad \checkmark$$

EVEN AND ODD Functions

A function is even if it is symmetric with respect to the y-axis.

Algebraic Check:

$$f(-x) = f(x)$$

A function is odd if it is symmetric with respect to the origin.

Algebraic Check:

$$f(-x) = -f(x)$$

Examples

Directions: Determine algebraically if the function is even, odd, or neither. If even or odd, describe the symmetry.

7. $f(x) = x^3 - x$

$$f(-x) = (-x)^3 - (-x)$$

$$= -x^3 + x$$

Odd; sym. about origin

8. $f(x) = -x^2 + 6$

$$f(-x) = -(-x)^2 + 6$$

$$= -x^2 + 6$$

Even; sym. about y-axis

9. $f(x) = x^3 - x^2 - 2x$

$$f(-x) = (-x)^3 - (-x)^2 - 2(-x)$$

$$= -x^3 - x^2 + 2x$$

None

10. $f(x) = \sqrt{x^2 - 4}$

$$f(-x) = \sqrt{(-x)^2 - 4}$$

$$= \sqrt{x^2 - 4}$$

Even; sym. about y-axis

11. $f(x) = \frac{-8}{x}$

$$f(-x) = \frac{-8}{-x} = \frac{8}{x}$$

Odd; sym. about origin

12. $f(x) = |x+1|$

$$f(-x) = |-x+1|$$

None